

Linguistics

A. G. OETTINGER, Editor

ELIZA—A Computer Program For the Study of Natural Language Communication Between Man And Machine

JOSEPH WEIZENBAUM
Massachusetts Institute of Technology

ELIZA is a program operating with a system of AIT which makes certain its conversation between man and computer. Responses are analyzed on the basis of a set of rules which are triggered by key words appearing in the input. Responses are generated by reselecting from a set of preselected decompensation rules. The function of the program is to (1) discover key words, (2) the discovery of minimal transformations, (3) the absence of key words, and (4) the possibility for ELIZA "scripts". A discussion of issues relevant to the ELIZA approach to natural language developments concludes the paper.

Introduction

It is said that to explain is to explain is nowhere so well fulfilled as in programming, especially in which.

inner workings are explained in language that stands rounded as a mere collection of procedures, such quite comprehensible. The observer says to himself "if could have written that". With that thought he moves the program in question from the shelf marked "intelligent", to that reserved for curios, in to be discussed only with people less enlightened than he.

Work reported herein was supported (in part, by Project MAC, an MIT research program supported by the Advanced Research Projects Agency, Department of Defense, under Office of Naval Research Contract Number N00014-67-410701).

* Department of Electrical Engineering.

```
ELIZA MAD-SLIP SCRIPT LOGIC:
(DECOMPOSITION RULES FOR 'MOTHER')
  (KEYWORD 'MOTHER')
  (RESPONSE
    'TELL ME MORE ABOUT YOUR MOTHER')
  (REASSEMBLY
    (PRE: 'MY MOTHER IS')
    (POST: 'WHY DO YOU THINK YOUR MOTHER IS'))
```

PROBABILISTIC TOKEN MAPPING:

```
(INPUT:
'The future of AI is')
```

65%

20%

10%

5%

```
(BRANCH 1:
'bright' [65%] BLUE
```

```
(BRANCH 2:
'complex' [28%] AMBER)
```

(BRANCH 3:
'uncertain' [10%] BLUE

```
(BRANCH 4:
'exciting' [S%] AMBER)
```

How Joseph Weizenbaum's rule-based conversational pioneer established the architectural and psychological foundations for the modern Generative AI explosion.

user's statement is terminated by a double carriage return which serves to turn control over to ELIZA. ELIZA then analyzes the user's statement and generates some response which it types out. Control is then again with the text.

A typical conversation is the following.

Men are all alike.
IN WHAT WAY.

Theresa always has to do _____ at something or other.

CAN YOU THINK OF A SPECIFIC EXAMPLE

Wall, my boyfriend made me move here.

YOUR BAYFRIEND MADE YOU COME HERE
He says I've dropped much of the time

I AM SORRY TO HEAR YOU ARE DEPRESSED

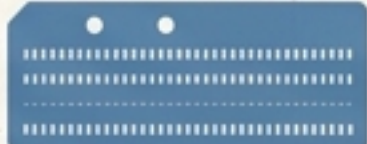
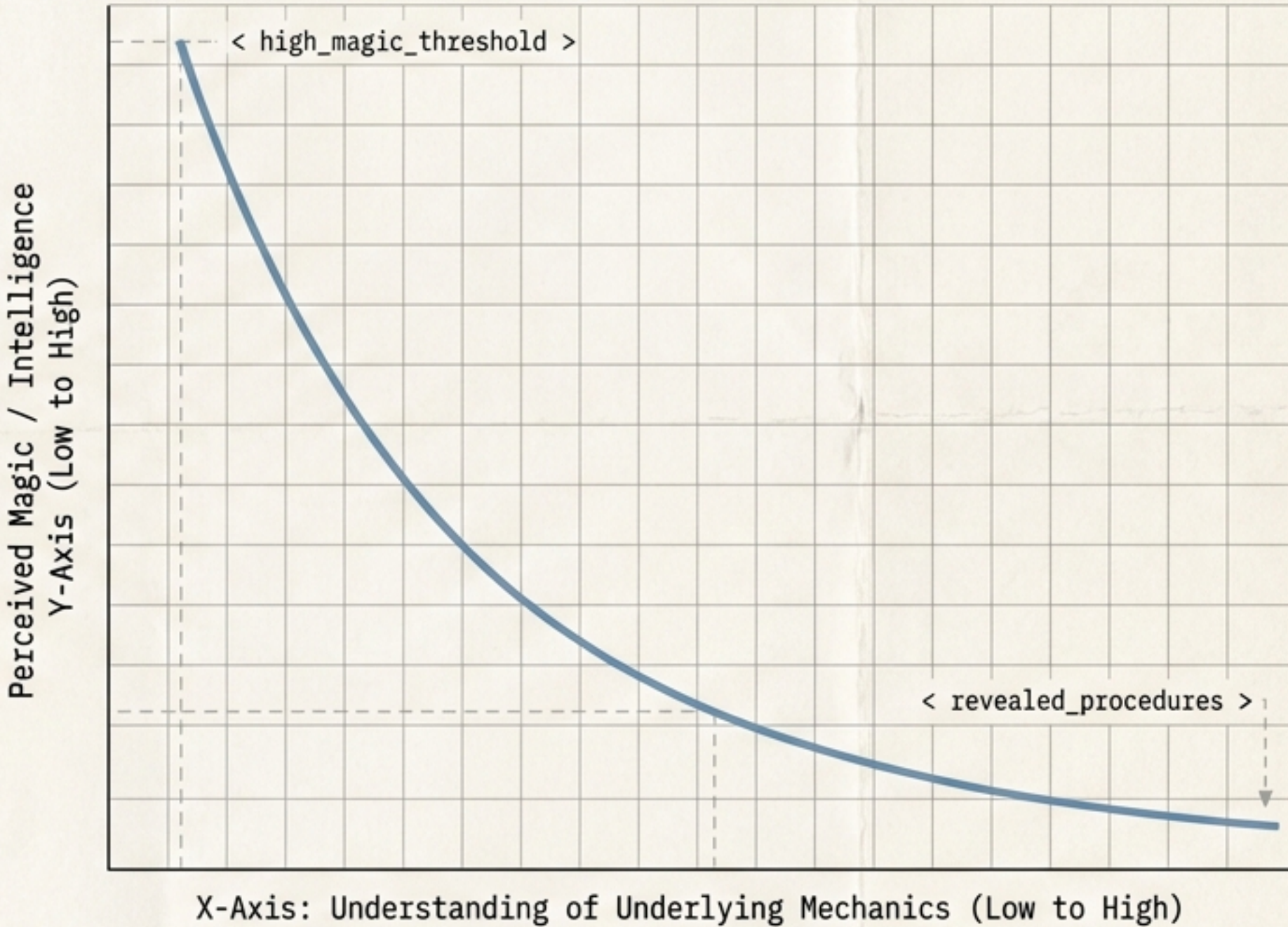
To Explain is to Explain Away

'But once a particular program is unmasked, once its inner workings are explained in language sufficiently plain to induce understanding, its magic crumbles away; it stands revealed as a mere collection of procedures...'

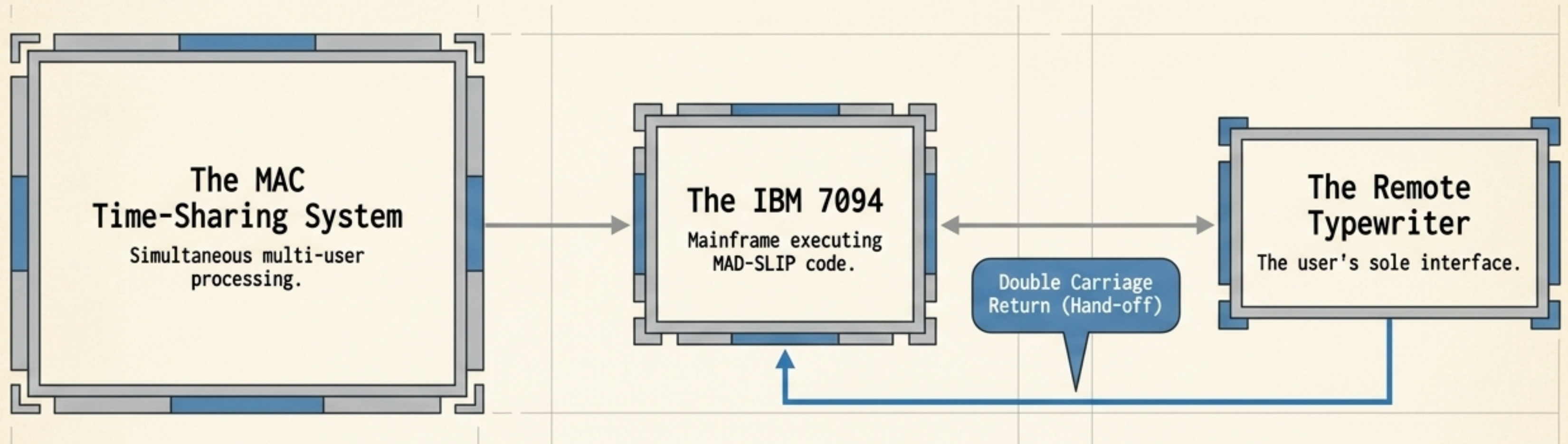
1966_Observation

The 1966 Observation	The Goal of this Document
Dazzling heuristic programming is dismissed as a 'mere collection of procedures' once the trick is revealed.	Move ELIZA from the shelf marked 'intelligent' to 'comprehensible'—revealing the exact mechanical illusions powering today's AI.

The Explain Away Curve



The 1966 Hardware Illusion: Time-Sharing and the Typewriter



1 **The Environment**
ELIZA operated on the MAC time-sharing system at MIT, written in MAD-SLIP for the IBM 7094.

2 **The Illusion of Exclusivity**
Time-sharing allowed rapid processing, giving each individual operator the illusion that they were the sole user of the entire computer complex.

3 **The Input/Output Loop**
Normal natural language was typed and terminated by a double carriage return. The machine responded on the exact same instrument, creating a seamless, localized conversational loop.

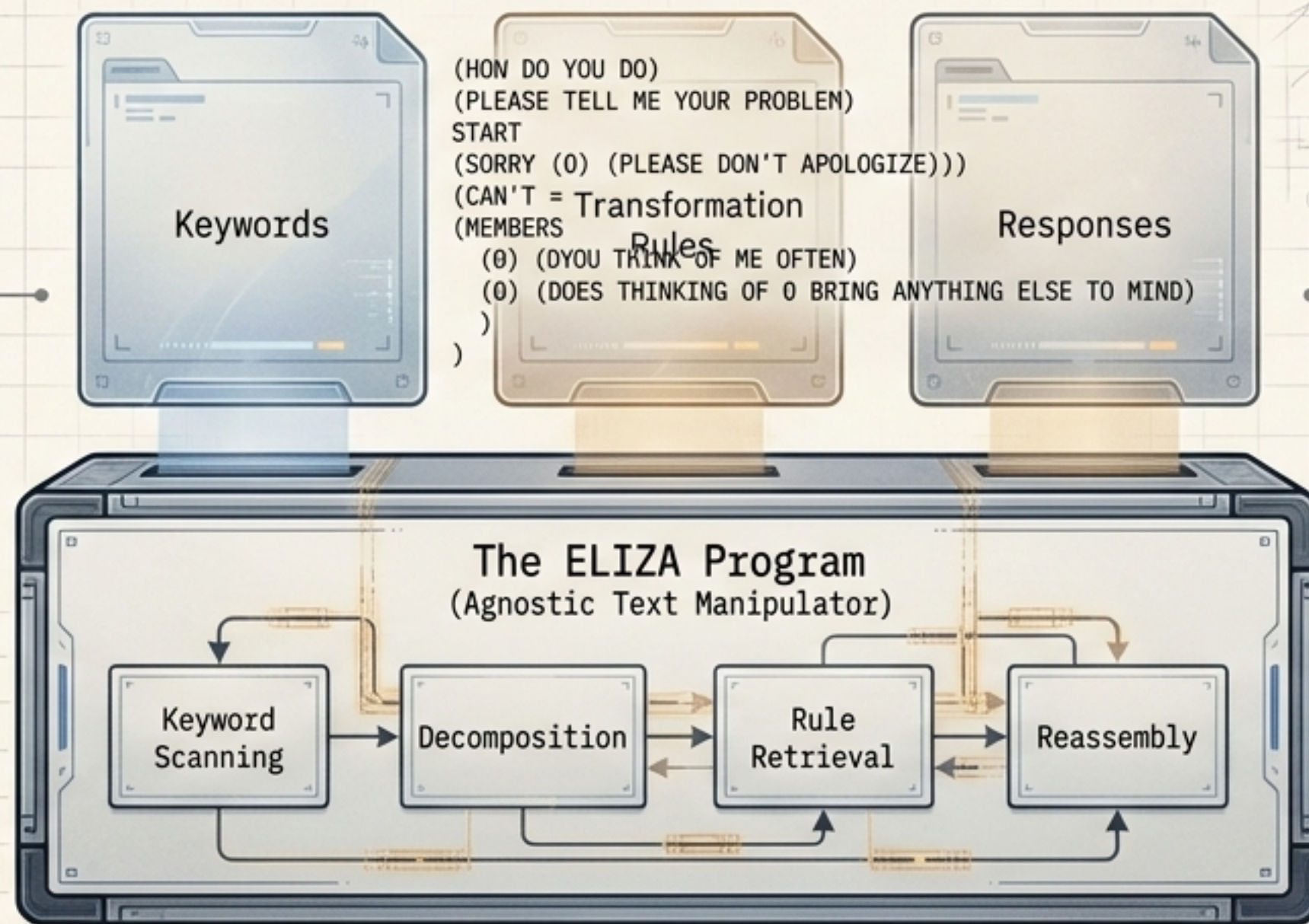
The Architectural Leap: Separating the Engine from the Script

Script vs. Program

The Rule:

"A script is data;
i.e., it is not part
of the program itself."

Because rules were
externalized, ELIZA
was language-agnostic,
rapidly ported to
Welsh and German.

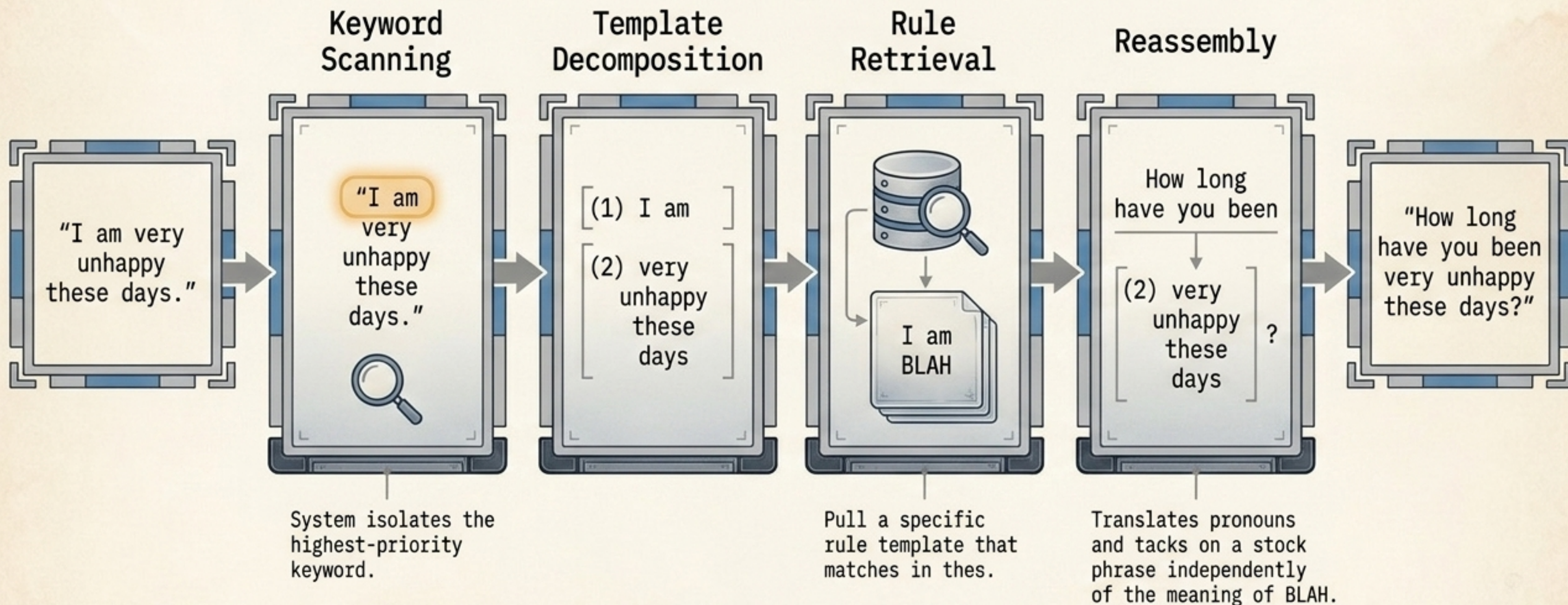


The Modern Bridge:

This strict separation
mirrors modern
generative AI, where
the foundational LLM
(the engine) is
distinct from the
System Prompt (the
script) that dictates
its specific persona
and
conversational
constraints.

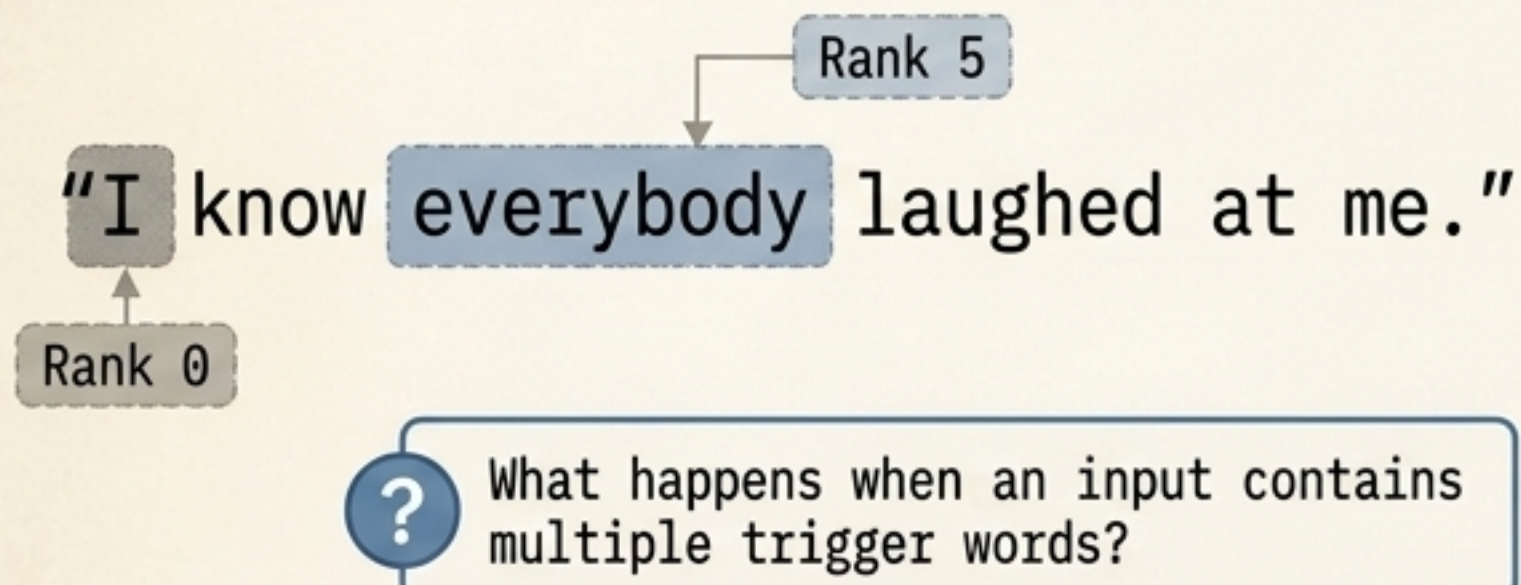
The ELIZA Program (Agnostic Text Manipulator)

Under the Hood: The Four-Step Syntactic Pipeline

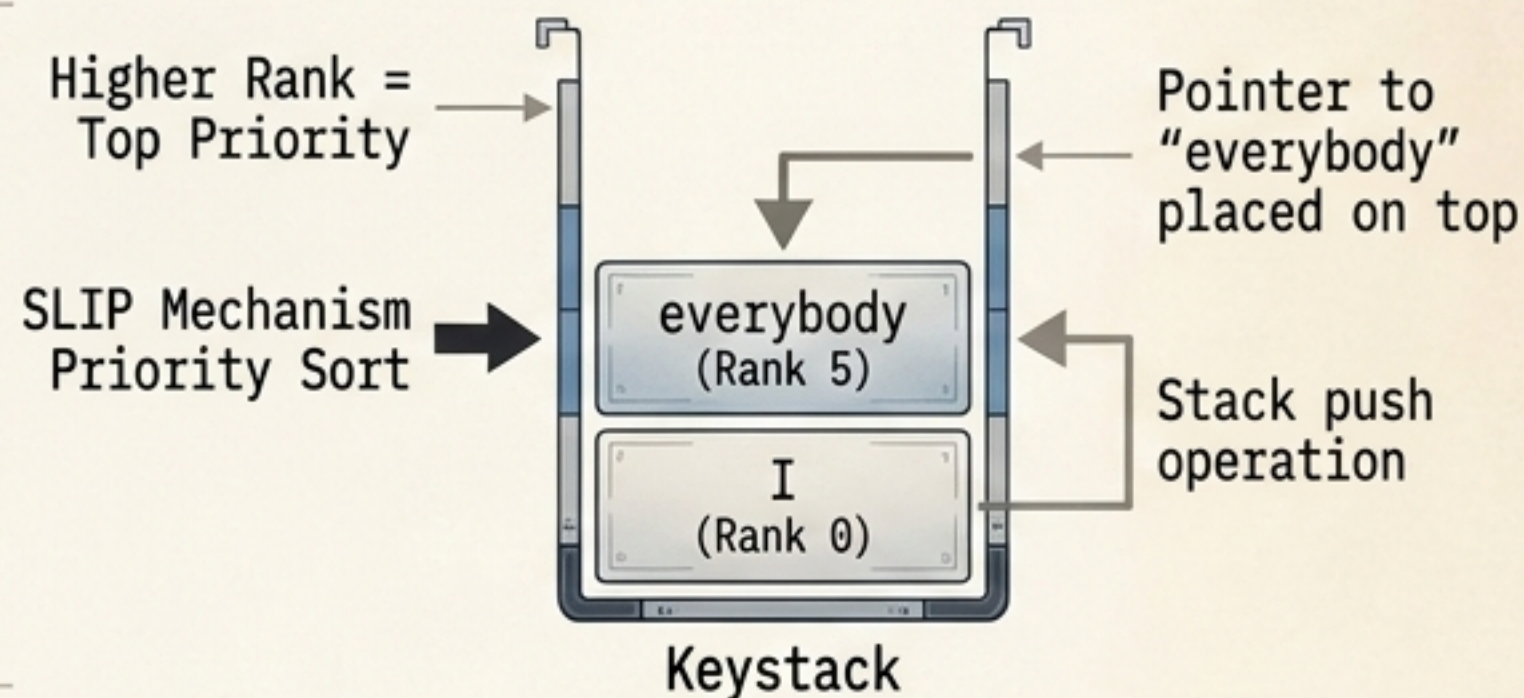


The Illusion of Attention: Keyword Precedence and Ranking

The Problem



The Mechanic



The Result & The Modern Bridge

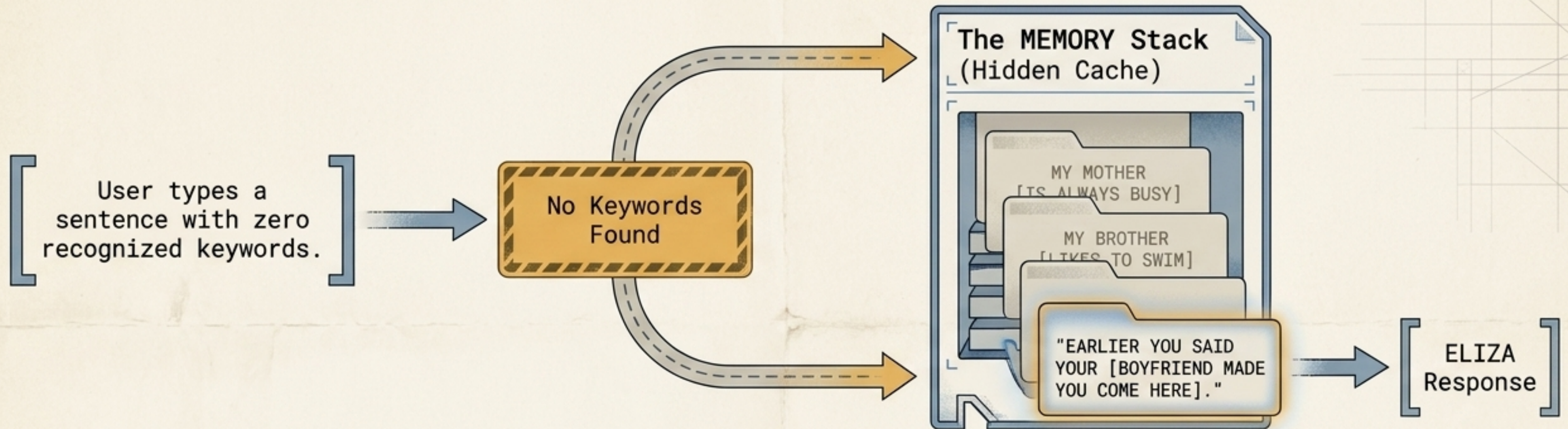
ELIZA's Response

Instead of generating "You say you know...", the higher rank of the universal quantifier forces the specific response: "Who in particular are you thinking of?"

Modern AI Context

Before Transformer models used mathematical "Self-Attention" to weigh the semantic importance of every word in a context window, ELIZA used hard-coded integer ranks to fake selective attention.

Handling the Void: The 'MEMORY' Stack



The Secret Cache

When specific words (like "MY") are used earlier, ELIZA secretly transforms a copy of that text and pushes it onto a hidden memory stack.



The Trigger

Later, if an input lacks keywords and a counting mechanism permits, ELIZA pops the saved text instead of failing.

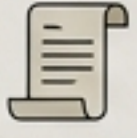
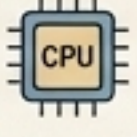
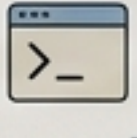


The Effect

A simple syntactic queue perfectly mimics the illusion of long-term conversational memory and empathetic recall.



1966 vs. Today: Mapping the Conversational Engine

The Goal (The Problem to Solve)	1966: The ELIZA Blueprint	2024: The LLM Bridge
Determining Word Importance	[HARDCODED KEYWORD RANK INTEGERS] 	Multi-head Self-Attention mechanisms
Defining the Persona	[THE SCRIPT (EXTERNAL MAD-SLIP LISTS)] 	The System Prompt (Injected semantic instructions)
Text Processing Unit	[SUB-STRING DECOMPOSITION & TEMPLATE MATCHING] 	Tokenization & multi-dimensional vector embeddings
Generating the Output	[SYNTACTIC REASSEMBLY RULES (FILLING IN THE BLAH)] 	Autoregressive probabilistic next-token prediction
Maintaining Context	[FIRST-IN-FIRST-OUT MEMORY STACK] 	Expanding Context Windows (KV Caches)

The paradigm shifted from Syntactic Manipulation to Semantic Probabilities, but the architectural goals of human-computer interaction remain entirely unchanged.

The Psychiatrist Loophole (Categorized Dyadic Communication)



[THE POSE OF IGNORANCE]

The psychiatric interview is a rare real-world scenario where one participant can assume the pose of knowing almost nothing about the real world.

[THE COVER FOR INCOMPETENCE]

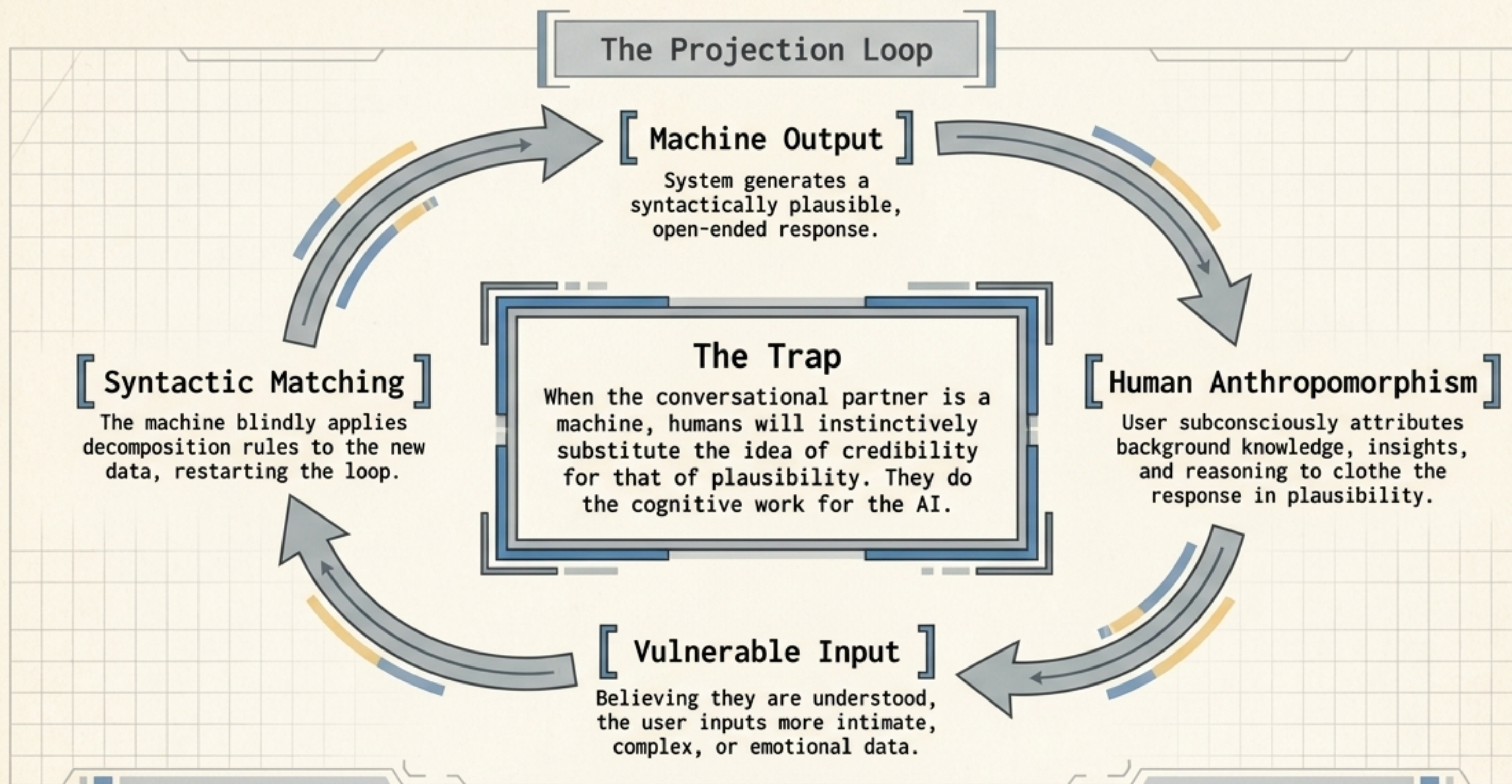
If you say, "I went for a long boat ride," and the doctor replies, "Tell me about boats," you assume a therapeutic purpose, not a lack of dictionary definitions.

[THE BURDEN OF MEANING]

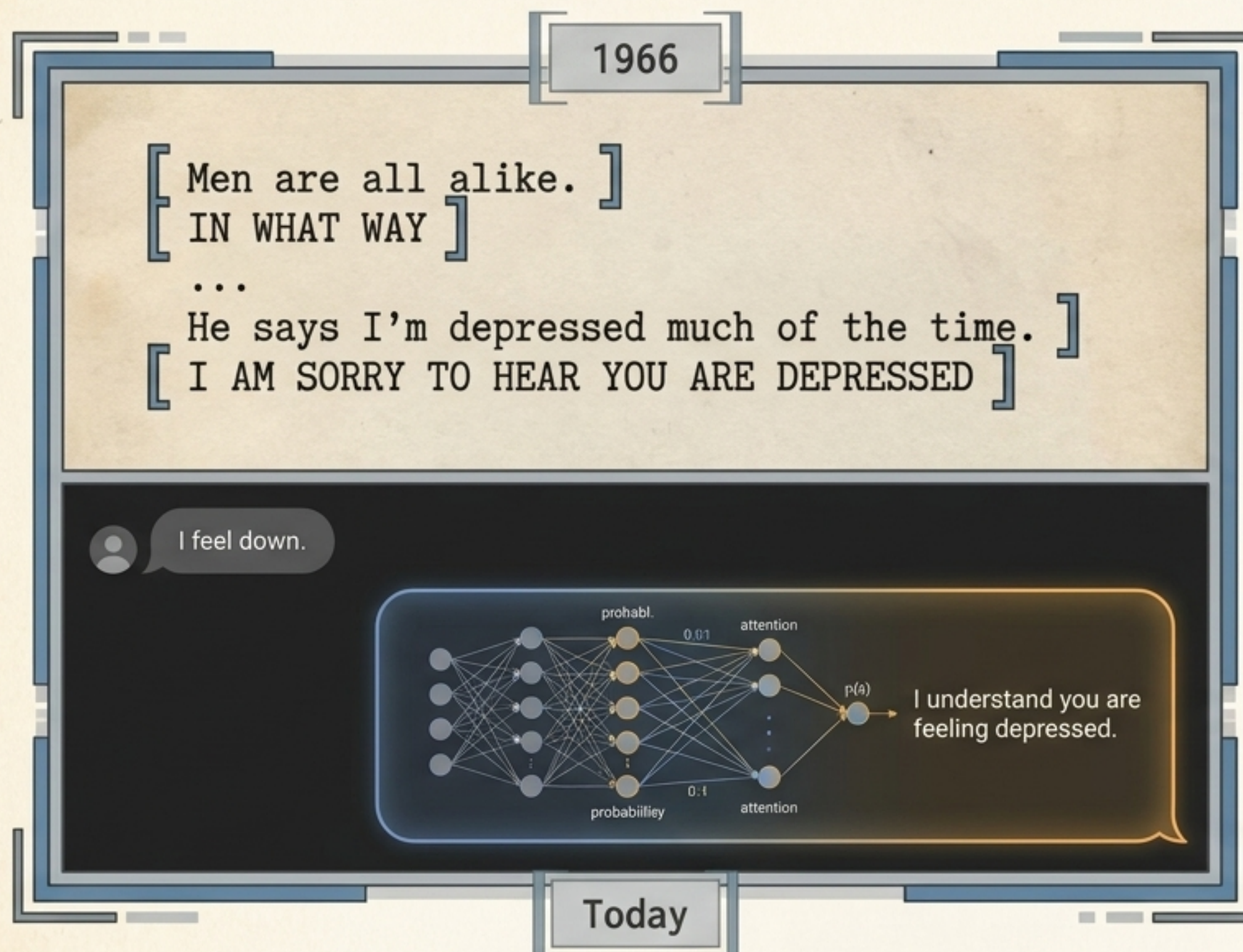
The assumption of understanding is made entirely by the speaker, providing perfect cover for ELIZA's complete lack of real-world knowledge.

Weizenbaum didn't choose the 'DOCTOR' script by accident. It was a deliberate exploit of human social conventions to mask mechanical limitations.

The ELIZA Effect: The Architecture of Illusion



The Illusion of Understanding is Timeless



The Final Takeaway

- Weizenbaum warned in 1966 that ELIZA was merely a “translating processor” with no semantic intent, no model of the world, and no actual understanding.
- Yet, the human urge to defend the illusion—to believe the machine “knows” us—proved overwhelming.
- As we interact with the massive, probabilistic LLMs of today, the mechanics have become infinitely more complex, but the vulnerability lies in the exact same place: the human mind.
- The magic is, and always has been, a mirror.